

Sessional test II
CS 602/CO434 Image Processing
19 October, 2020
Marks: 20 Time: 30 minutes

1. Briefly answer the following:
 - a) Differentiate between point processing and filtering. 2
 - b) Name two mathematical operators that are used in image sharpening. 2
 - c) State the meaning of the following representation: 2
$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$
 - d) List the names of three smoothing filters and three sharpening filters. 2
 - e) Explain the gamma-correction process. 2
2. For the image given below perform the following operations.

2	8	1	6
2	5	3	9
4	0	6	2

- a) Smooth the image applying the 3x3 binomial filter. Show the result for the central two locations of the image only. 2
 - b) Sharpen the image using any one of the Laplacian filters. Show the result for the central two locations of the image only. Present the Laplacian filter used. 3
3. Provide to the point short answer: 1×5=5
 - a) Which one of the two filters - binomial and median causes more blurring?
 - b) Name the image sharpening technique used by newspaper industry?
 - c) Present the filter mask for one dimensional binomial filter of size 5.
 - d) Fill up the blank in the following:
Histogram _____ generates a processed image that has a specified histogram.
 - e) State whether the following statement is true or false:
Sum of weights in smoothing filters is always zero and sum of weights in sharpening filters is always one.